Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/23/2005
PI Name:	Stone, Jennifer Ph.D.		
Project Title:	The role of the homeobox transcription factor cProx1 in the development of the vestibular periphery in chickens		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
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No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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	The vestibular system coordinates motor responses to changes in head position and movement and is adaptable to different gravitational conditions. These features make vestibular function critical for space travel. Vestibular function depends upon peripheral receptors (hair cells) located in five inner ear organs and the nearby VIIIth cranial ganglion neurons that connect hair cells to the brain. These cells form during embryogenesis from the otocyst. There is evidence that development of the vestibular periphery is affected by space travel. Elucidation of the factors mediating microgravity-induced changes in vestibular development requires an understanding of normal inner ear formation. However, many aspects of normal otocyst development, including how the sensorineural regions become specified and acquire their normal cellular organization, remain unclear. The proposed experiments will examine the role of the homeodomain transcription factor, cProx1, in development of the		

Task Description:	vestibular epithelium and ganglion in normal chickens. This factor, which is critical for development of the lens, lymphatic system, and liver in mice, is highly expressed in the chicken otocyst in regions that will form the vestibular epithelia and ganglia. cProx1 expression is down-regulated after hair cells have formed, and it is re-expressed in mature chickens during hair cell regeneration. We will characterize temporospatial patterns of cProx1 mRNA and protein expression in the developing otocyst using in situ hybridization and immunofluorescence. In addition, cProx1 gene expression will be perturbed in ovo to test its role in sensory specification of the otocyst and in cell proliferation, differentiation, and patterning within sensory patches. Finally, the role of the hindbrain and periotic mesoderm in regulating the localized expression of cProx1 will be examined by transplanting otocysts into the hindbrain and by rotating otocysts in situ relative to the hindbrain. This research aims to expand upon our growing understanding of the molecular interactions necessary for vestibular end organ development in vertebrates.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The peripheral vestibular system is responsible for sensing head position and movement and is therefore critical for comfort and survival on Earth as well as in space. The goal of my research is to understand how the vestibular end organs and neurons are generated during development and regeneration in the chicken. This series of studies investigates the role of a homeobox transcription factor, cProx1, in the specification of sensory and neural cells within the developing inner ear. Identification of regulatory molecules that are necessary for otocyst morphogenesis and cellular patterning will lead to a better understanding of normal development and will help us unravel the cellular mechanisms underlying vestibular dysfunction.
Task Progress:	The goal of this project was to explore the expression and the role of the divergent homeobox transcription factor, Prox1, in development of vestibular hair-cell epithelia and ganglia. This year was a no-cost extension for this project. Three studies were completed. First, we further developed methods for misexpressing Prox1 in the embryonic chicken otocyst using electroporation and adenoviral gene delivery methods. Both methods lead to robust overexpression of nuclear Prox1 throughout the otocyst, including in vestibular epithelia and the cochleovestibular ganglion. We are currently analyzing phenotypes derived from misexpression experiments, in developing and regenerating vestibular epithelia. Second, we explored using immunofluorescence the expression of Prox1 protein during development of the mouse inner ear. Prox1 arises early in the primordial of vestibular epithelia and in the vestibular ganglion. Expression is seen transiently in post-mitotic hair cells, but it is sustained in supporting cells until around birth. Prox1 is also highly expressed in the organ of Corti. Expression is again transient in hair cells, being down-regulated prior to the emergence of MyosinVIIa immunoreactivity. However, expression in supporting cells is sustained until several days postnatally. A paper describing these results has been submitted and is under revision. Third, we completed characterization of the expression of four calcium binding proteins (calbindin, calretinin, parvalbumin, and calmodulin) in the developing and post-hatch vestibular periphery using immunolabeling. Calbindin and calretinin labeling indicates the location of developing sensory patches by E3.5, whereas labeling for calmodulin emerges at E5 and labeling for parvalbumin emerges at E14. All four markers show distinct sensory and neural labeling throughout the inner ear in the mature bird. A paper describing these results will be submitted in December, 2005.
Bibliography Type:	Description: (Last Updated: 09/19/2006)
Abstracts for Journals and Proceedings	Bermingham-McDonogh O, Stone JS, Hume C, Oesterle EC. "Developmental expression of the homeobox transcription factor Prox1 in the murine organ of Corti." Association for Research in Otolaryngology Midwinter Meeting, Feb 2005. Association for Research in Otolaryngology Midwinter Meeting, Feb 2005, vol 28: 283. , Feb-2005
Abstracts for Journals and Proceedings	Falkner A, Denenny K, Stone JS. "Immunoreactivity for calcium binding proteins in developing and mature sensory epithelia of the avian inner ear." Association for Research in Otolaryngology Abstracts, Feb 2004. Association for Research in Otolaryngology Abstracts, Feb 2004, 27. , Feb-2004
Articles in Peer-reviewed Journals	Bermingham-McDonogh O, Oesterle EC, Stone JS, Hume CR, Huynh HM, Hayashi T. "The expression of Prox1 during mouse cochlear development." J Comp Neurol. 2006 May 10;496(2):172-86. PMID: 16538679 , May-2006